A Hybrid Intrusion Detection System Utilizing BERT Contextual Log Parsing and Random Forest Classification for Real-Time Zero-Day Distributed Denial of Service Mitigation

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Introduction

Research Topic

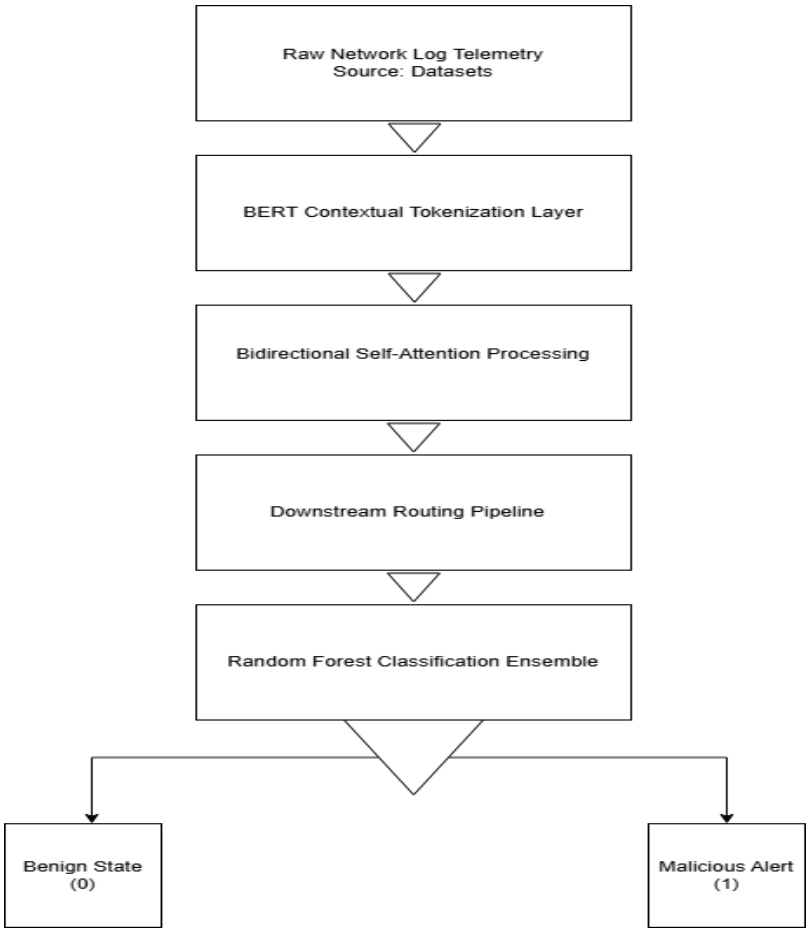
Modern enterprise entities, regardless of their size, generate a large, continuous flow of telemetry data. In these environments, attacks such as Distributed Denial of Service (DDoS) are common and cause service disruptions. These types of attacks use massive volumes of coordinated traffic to overwhelm the target infrastructure, leading to either slower application speeds for users or a complete loss of service. To combat these attacks, most enterprises use traditional defensive mechanisms such as Signature-Based Intrusion Detection Systems (IDSs), in which systems evaluate and compare incoming traffic against a set of preconfigured threat- and non-threat-labeled data stored in their databases. While these systems are very efficient in terms of compute resource costs and line-rate execution, predetermined pattern-matching models suffer from a vulnerability that's fundamental to their structure. They are entirely unable to identify zero-day vulnerabilities and polymorphic attacks that have yet to be established as threat signatures in the database.

To address this limitation of the IDS system, this research introduces a hybrid intrusion detection framework that uses deep semantic analysis and body machine learning as a lens for analyzing network log telemetry. The core research topic focuses on the development, optimization, and empirical evaluation of a sequential two-stage Machine Learning pipeline. This system integrates Bidirectional Encoder Representations from Transformers (BERT) for context-aware log parsing and structural feature extraction, along with an optimized downstream Random Forest classification engine for real-time threat analysis and mitigation.

The mechanics behind this architecture differ from those of traditional machine learning models, which depend on static tabular engineering. Instead of treating log attributes as isolated numerical variables, this system implements a translation pipeline that serializes multi-dimensional telemetry metadata into structured, linguistic sentences. Network fields such as destination ports, transmission protocols, active flow durations, and even packet count multipliers are directly mapped into a natural text syntax. This form of textual transformation allows the framework to exploit the grammar hidden in machine-generated communications.

Once arranged together, these narrated log strings are processed by a pre-trained BERT transformer engine. Using a directional self-attention mechanism, the transformer layer tokenizes the log strings and maps the linguistic relationships between distinct system fields into

a high-dimensional vector space. This operation extracts dense, sequential semantic feature embeddings that capture the context of changes in network state.



To mitigate the inherent high computational overhead of transformer models, this system routes the extracted embeddings directly to a downstream Random Forest Classifier. This secondary layer serves as an engine for rapidly evaluating information. It uses a parallel combination of independent decision trees to perform fast decision splits on the dense BERT matrices.

The validation and training segments of this hybrid pipeline were based on multi-generational network traffic repositories, which drew data from both the CICIDS2017 and CSE-CIC-IDS2018 datasets. By combining historical slow-rate protocol exploitations with modern high-volume application-layer flood variants, the framework built a highly versatile definition of network anomalies. This specific architectural synthesis provided a defense pipeline capable of interpreting the structural context of network events while maintaining the fast execution speeds required for active production environments.

Importance

The current condition of most, if not all, digital operations globally is constrained by complete dependence on high-availability cloud platforms, distributed financial systems, and highly critical industrial network frameworks. As the size of these systems increases, so does

their complexity; they're under a constantly rising threat from advanced, uncategorized cyber threats that affect their ability to serve their users. For example, Zero-day DDoS attacks represent an especially dangerous class of attack since they exploit undiscovered protocol weaknesses or application flaws before defensive security developers can write or deploy relevant patches. When corporate security operations rely very exclusively on reactive tools, they create a gap in the system's defense. Then, this gap leaves the infrastructure fully exposed during the critical time window between the live weaponization of a zero-day exploit and the eventual distribution of a corresponding rule update.

Furthermore, the exponential increase in enterprise data volumes has rendered traditional signature-based machines very obsolete. High-throughput gateways under heavy loads face processing bottlenecks when forced to parse packet payloads while their rules are constantly expanding. This constraint leads to frequent dropped packets or the bypassing of forced inspection.

At the same time, traditional rule-of-thumb or statistical anomaly-detection methods frequently suffer from higher false-positive rates. When exposed to standard and non-linear operational shifts in dynamic networks, simple rule-based anomaly engines frequently misclassify legitimate structural changes as active threats. This constant stream of inconsistent alerts causes widespread alert fatigue within Security Operations Centers (SOCs). When Human security analysts become overwhelmed by the legitimate volume of false alerts, their time to respond degrades. This increases the likelihood that a legitimate, low-rate intrusion will be overlooked amid all the false system noise.

This research project is highly important because it addresses the concurrent requirements of contextual depth and processing speed in information security monitoring systems. By treating each network telemetry event as an active language syntax, the BERT parsing layer matches subtle anomalies based on contextual disruptions rather than rigid definitions. The model can isolate malicious intent even when an attacker changes packet headers or obfuscates payload signatures to bypass standard rule-based systems.

At the same time, tasking the final classification choice to an optimized Random Forest model solves the latency limitations that usually prevent transformer architectures from being implemented and deployed at the boundaries of active networks. The practical viability of this approach is demonstrated by its performance metrics, achieving an overall accuracy of 83.80% on a multi-dataset validation set of 4,500 instances. This empirical finding demonstrates that a hybrid pipeline can successfully maintain a clean, resilient true-negative filtering baseline while detecting highly complex, signatureless anomalies.

Background & Literature Review

Summary

The use of Artificial Intelligence to monitor and protect network systems has evolved through distinct stages of technology, essentially mirroring broader progress in data-processing hardware. Early research in automated network intrusion systems focused primarily on basic machine learning algorithms. These early frameworks relied heavily on methods such as single decision trees, support vector machines, and simple Naïve Bayes Classifiers. These legacy industry

frameworks required extensive manual feature engineering. Human domain experts had to identify, isolate, and calculate relative network attributes. For example, attributes such as inter-arrival times or specific flag distributions can be used before exposing the data to the matrix for classification. The manual approach took an extreme amount of time and was prone to human bias. Because Human operators often failed to map the multidimensional levels of relationships that make up the identity of an advanced cyber-attack.

The introduction of deep learning frameworks has made efforts to automate this feature extraction process. Frameworks and Architectures such as Long Short-Term Memory Networks and Convolutional Neural Networks demonstrated a strong theoretical ability to recognize hidden structures in massive data streams during controlled laboratory tests and trials. They achieved high accuracy scores but introduced significant operational strain when transitioned to live production environments. Their increasingly high computational complexity required substantial graphics processing hardware; one example is NVIDIA Blackwell GPUs. Eventually, creating a processing-speed delay that severely degraded network performance on regular enterprise gateways. Additionally, deep neural networks function completely as uninterpretable black boxes. They spit out binary classification options without providing any context or explaining why the information was flagged as an anomaly. This lack of transparency slows incident response workflows, since analysts must manually audit raw hexadecimal data to identify and verify the system's automated outputs.

To solve these visibility and efficiency constraints, modern security literature has turned towards NLP (Natural Language Processing) Architectures for analyzing system-generated logs. Machine-generated system messages follow a distinct, structured grammar that closely mirrors human language. When an exploit occurs, it actively changes the vocabulary, sequence, and context of these log entries. Pre-trained transformer models can be fine-tuned to learn the standard baseline behaviors of an infrastructure. When a threat enters the system, the transformer would then identify a break in the semantic language continuation. It will flag the entry as an anomaly based on the structure of its context, rather than relying solely on a strict match of the signature.

The current consensus in cybersecurity research prioritizes the development of hybrid models. These systems pair up the deep semantic intelligence of the transformers with the rapid execution speeds of shallow ensemble models. These sequential designs aim to develop dependable, adaptable security systems capable of protecting high-speed production networks against emerging threats and vulnerabilities.

Key Studies

The theoretical and empirical (yes, I apologize, this is a new buzzword I learned and have spammed quite a bit) structure of this proposal is built on top of specific investigations spanning language modeling, ensemble learning, systems optimization and zero-day threat analysis.

The use of language models for proactive threat hunting is supported by research by Al-Shari (2025), who introduced the ZeroDay-LLM framework. This study demonstrates that large language models can analyze the underlying context of network data to discern previously

unknown attack patterns with higher statistical accuracy. By training the architectures to evaluate the semantic meaning of packet structures rather than relying solely on static numeric and boolean definitions, the ZeroDay-LLM framework successfully minimizes false positives across diverse environment benchmarks. This research provides us with an empirical (used it again) justification for treating network telemetry as a text string, showing that evaluation is a highly viable alternative to traditional rule-based signature engines.

Chen and Rao (2024) conducted an exploration of the specific application of transformer architectures as a self-supervised system for parsing logs and verifying their sequences. Their research analyzed the mechanisms at the core of the encoder, detailing how self-attention weights enable the system to perform log parsing and sequential verification. They demonstrated that, through targeted training tasks, transformers effectively learn the standard dependencies of system log structures across massive infrastructure datasets. The research in this study provides a technical blueprint for the log parsing phase of this proposal, detailing how attention layers isolate subtle formatting shifts to flag anomalies.

While Transformer models provide deep contextual understanding, their execution requires pairing them with an efficient classifier to meet real-time processing speed constraints. Alfariz and Khair (2025) provided us with a very practical evaluation of the Random Forest Algorithm for real-time attack detection in web hosting environments. Their work emphasizes the structure of a Random Forest Model, which uses an ensemble of independent decision trees, enabling low-latency processing and immediate inference times. They highlighted the absolute necessity of proper feature selection, demonstrating that when a model targets the most information-rich log parameters, it can trigger automated defensive actions instantly without lagging behind the underlying server infrastructure.

The requirement for high-speed execution at the system level is examined further by Bugden(2022) in an analysis of systems programming languages for critical security architectures. Bugden provided extensive performance benchmarks showing that modern memory-safe languages achieve processing speeds equivalent to legacy-grade languages like C and C++, while removing common vulnerabilities such as buffer overflows. This work provisions(I don’t wanna say provide for the billionth time) us with a strong technical justification for why a high-performance network security framework benefits from optimized data ingestion pipelines, ensuring that the components that feed the machine learning models do not end up bottlenecked under conditions of high traffic volumes.

To address the elephant in the room: the challenge of fully undocumented attack vectors. Al-Abassi (2024) developed a specialized framework for zero-shot learning to identify cyber-attacks that are completely absent from the initial training dataset. When tested in a dynamic vehicular network environment, the study showed that by mapping specific semantic knowledge into a deep neural network, an authentication engine can successfully identify new malware variations without requiring immediate retraining. This study sheds light on a significant expansion goal for this proposal: the long-term scalability of the semantic architecture.

The validity of the architecture, which combines multiple distinct machine learning approaches, is confirmed by Rabiu (2025) in a comprehensive and systematic review of hybrid intrusion detection systems. Rabiu evaluated peer-reviewed studies to analyze the performance of models with multi-stage architectures, concluding that architectures pairing an unsupervised anomaly detection layer with a supervised classification layer consistently outperform single-model frameworks. This review noted that hybrid designs are significantly more resilient when pitted against polymorphic malware and zero-day exploits because they actively separate the task of structural feature extraction from the choice of final classification.

The practical deployment of transformers for log file evaluation was pioneered by the one and only Chen (2022) through the development of the BERT-Log framework. This study treated the system messages as a natural-language body, passing them through an encoder before using a fully connected layer in a neural network to fine-tune anomaly scores. The model achieved exceptionally high precision scores on open-source datasets, fully outperforming traditional sequence-based models. Chen's detailed breakdown of the window-sliding approach provides the foundational mathematical logic for the data preprocessing phase of this research.

To ensure the ensemble classification layer operates at full efficiency, Zhou (2025) investigated feature optimization techniques for machine learning-based models tasked with DDoS mitigation. Zhou compared multiple supervised classifiers, concluding that a feature-optimized Random Forest model provides a more dependable balance between pure raw accuracy and rapid inference speeds. The study shed light on specific feature-selection metrics that directly inform the parameter selection for this project.

The underlying Security crisis that's driving the momentum of this architectural research is outlined by Hindy (2023) in a broad survey of machine learning applications for zero-day threat detection. HIndy presented operational data showing that zero-day vulnerabilities frequently go undetected within company networks for many days before being discovered. Pressing emphasis on the severe operational limitations of the legacy systems. The survey also discusses the critical role of explainable AI systems, noting that when a model can provide transparent anomaly alerts, analysts can intercept threats much faster.

Finally, the low-level technical execution of security tools requires a strong, strict focus on code integrity, a concept Sharma (2025) investigated through targeted fuzzing of high-performance code blocks. Sharma noted that low-level interactions require ingesting raw network packets and often require code blocks that bypass default compiler checks.

Gaps in Current Literature

Despite the extensive detail in the current security literature on progress, several major gaps remain between theoretical models and the realities of operational deployments. The most important gap involves the extreme computational trade-off between semantic depth and processing latency. While current studies show that transformer frameworks achieve high accuracy in parsing context, these architectures incur significant computational overhead. Existing research ignores the constraints of the physical hardware in high-throughput network

gateways, where thousands of packets must be analyzed every second. There is a documented lack of empirical research (I got it, use it again) on integrating a deep transformer layer with a lightweight ensemble model to create a unified pipeline that maintains low processing latency without sacrificing context accuracy.

Another limitation in the existing body of knowledge concerns authentic data and realistic environments. A significant majority of machine learning research in IDS relies on outdated or highly stylized synthetic datasets. These simulated datasets are often scrubbed clean of the background noise, formatting errors, and erratic behaviors that characterize real-world enterprise networks. Consequently, models achieving near-perfect accuracy in academic publications frequently experience significant performance degradation when deployed in live production network environments. This research proposal directly addresses these gaps by evaluating a hybrid architecture designed to balance semantic capabilities with rapid execution speed, using realistic multi-generational datasets to provide better insights.

Method and Design

This study uses an experimental and comparative design methodology. To fully evaluate the performance of the proposed hybrid architecture, the experimental setup compares an experimental group against a clearly defined control group. The experimental group consists of the hybrid intrusion detection pipeline. This system uses a pre-trained language model for semantic log parsing and feature extraction, combined with an optimized Random Forest classifier for final traffic categorization. The control group consists of a standard, industry-recognized signature-based IDS configured with up-to-date core rule sets.

The primary objective of this experimental design is to isolate the detection system's architectural configuration as the independent variable. The dependent variables are the system's defensive efficacy and accuracy against identical traffic inputs. By establishing a strictly controlled virtualized testing environment, the study ensures that external factors such as network hardware variety or fluctuating host CPU workloads are largely minimized. Both defensive systems are exposed to zero-day simulation vectors. This comparative layout provides a transparent method for measuring whether the semantic parsing layer enables the hybrid model to detect novel threats that bypass rule-based defenses entirely.

Data Collection and Stratification.

Data collection is performed via an automated pipeline in an isolated laboratory environment. Deploying experimental scripts to a live company network introduces packet loss and unmappable user behavior, compromising control of the experimental environment. Therefore, the study combines packet data sources from the CICIDS2017 and CSC-CIC-IDS2018 datasets. Which were developed by the Canadian Institute for Cyber Security, hold a widely accepted set of industry benchmarks, which contain realistic profiles for diverse cyber attacks running alongside normal baseline network behaviors produced by a dedicated user profile system.

Merging the 2017 and 2019 datasets is a critical architectural design choice. Relying on a single year of data can severely limit a model's exposure to different traffic distribution backgrounds. The CSE-CIC-IDS2018 dataset focuses strongly on high volumes of application-layer floods. Specifically, Low Orbit Ion Cannon and High Orbit Ion Cannon attack vectors. On the other hand, the CICIDS2017 dataset brings in a slower structural connection-exhaustion profile, such as Slowloris and Goldeneye. Combining them creates a highly complex data pool, testing the model's ability to detect stealthy anomalies hidden within massive volumes of company data.

To begin the experiment, a large sample size of network log instances must be extracted from the combined dataset utilizing a stratified random selection process. The extraction method targets a blended validation slice of exactly 4,500 specific instances. This technique ensures that the sample proportionally represents all major traffic classifications, preventing the model from developing a classification bias toward overrepresented classes.

Because of the high memory consumption of the raw datasets, the ingestion pipeline uses the Parquet file format. Parquet files are compressed binary files that load much faster and use less memory than raw CSV files. The Python testing environment uses the PyArrow engine to stream these files into memory-safe chunks, preventing system resource exhaustion during the datasets' initial loading and randomization phases.

Tabular to Semantic Text Serialization

The foundational mechanism that's behind the proposed architecture is the text serialization process. Machine learning models traditionally ingest network traffic as rigid table vectors. This study pushes the data into a human-readable format to maintain an understandable narrative. The preprocessed scripts isolate specific network parameters, including destination port values, transmission protocol integers, flow duration measurements, and packet payload counts.

These isolated numbers are directly mapped into a standardized syntax template. A single row of network traffic is converted into an explicit string of text. The resulting sentence describes the target port, the active protocol, the exact microsecond duration of the connection, and the total inbound and outbound metrics.



```
all_files = os.listdir(DATA_DIRECTORY)
parquet_files = [os.path.join(DATA_DIRECTORY, f) for f in all_files if f.endswith('.parquet')]
csv_files = [os.path.join(DATA_DIRECTORY, f) for f in all_files if f.endswith('.csv')]

if not parquet_files and not csv_files:
    raise FileNotFoundError(f"No usable data logs discovered in '{DATA_DIRECTORY}'. Please verify your file placement.")

collected_frames = []

# 1. Process 2018 Parquet Files
for file_path in parquet_files:
    print(f"Ingesting 2018 Parquet file: {os.path.basename(file_path)}")
    df_chunk = pd.read_parquet(file_path, engine='pyarrow')
    df_chunk.columns = df_chunk.columns.str.strip()

    if 'Label' in df_chunk.columns:
        df_chunk = df_chunk.dropna(subset=['Label'])
        collected_frames.append(df_chunk)

# 2. Process 2017 CSV Files
for file_path in csv_files:
    print(f"Ingesting 2017 CSV file: {os.path.basename(file_path)}")
    for chunk in pd.read_csv(file_path, chunksize=50000, low_memory=False):
        chunk.columns = chunk.columns.str.strip()

        if 'Label' in chunk.columns:
            chunk['Label'] = chunk['Label'].str.strip()
            chunk['Label'] = chunk['Label'].replace({
                'BENIGN': 'Benign',
                'DDoS': 'Attack',
                'DoS Hulk': 'Attack',
                'DoS GoldenEye': 'Attack',
                'DoS slowloris': 'Attack'
            })
            chunk = chunk.dropna(subset=['Label'])
            collected_frames.append(chunk)

        current_pool = sum(len(df) for df in collected_frames)
        if current_pool >= TARGET_SAMPLE_SIZE * 3:
            break

master_expanded_df = pd.concat(collected_frames)
print(f"\nTotal multi-generation dataset successfully loaded: {len(master_expanded_df)} rows.")

sampling_ratio = TARGET_SAMPLE_SIZE / len(master_expanded_df)
stratified_subset, _ = train_test_split(
    master_expanded_df,
    train_size=sampling_ratio,
    stratify=master_expanded_df['Label'],
    random_state=42
)

print(f"\nStratified sample allocation complete. Total target size: {len(stratified_subset)} rows.")
print("Updated Training Profile Class Distribution:")
print(stratified_subset['Label'].value_counts())
```

This textual transformation closes the gap between our raw networking data and NPL. It enables attention in the transformer's deep layers to accurately capture the variables' structural dependencies, rather than just assigning mathematical weights to abstract numeric classes. Corrupt records, Missing labels, and half-written log strings are forcibly dropped during the serialization phase to maintain strict data integrity.

Contextual Embedding Pipeline

The serialized narrative log strings are processed by a Hugging Face transformer Infrastructure. The pipeline initializes the bert-bert-uncased pretrained model. The text arrays are then passed directly into a WordPiece tokenizer. This part of the component separates the complex string into recognizable sub-word units, applies the necessary padding tokens, and truncates the boundaries so that it's possible to maintain a constant maximum length of 64 tokens per sequence. (Quit a lot of buzz words there, more coming)

The core transformer engine then processes these token arrays. Using deep bidirectional self-attention, the architecture then computes the linguistic relationships between every token in the sequence. The mathematical Logic defining this mechanism is represented by using the scaled dot-product attention formula: $Attention(Q,\ K,\ V)\ =\ softmax(\frac{QK^t}{sqrt(d_k)})V$

The equation shows how the query vectors (Q) and the key vectors (K) are multiplied and scaled by the key dimension (d_k) before being multiplied by the final value weights (V). The embedding extraction function explicitly targets the high-dimensional hidden vector associated with the sequence's anchor classification token. This output matrix then condenses the semantic and structural context of the entire log sentence into a strict 768-dimensional mathematical array.

Since running a full transformer evaluation on CPU units requires a massive amount of computational power, the data arrays are strictly processed into controlled batches. The resulting context matrices are compiled and then split into training and testing subsets using a standard 70:30 allocation ratio, with a fixed random state to ensure experimental reproducibility.

Random Forest Classification and Hypothesis Testing

The high-dimensional feature matrices generated by the parser are routed directly to the RF classification Layer. The Random Forest algorithm constructs an ensemble of orthogonal decision boundaries by applying bootstrap aggregating to its training data. At each node split, a random subset of the 768 extracted features is evaluated to minimize the Gini impurity of the classification decision. The Gini impurity evaluates the probability of an incorrect classification of data based on the following mathematical representation:

$$Gini(p)\ =\ 1\ -\ \sum_{i=1}^{J} p^2_i$$

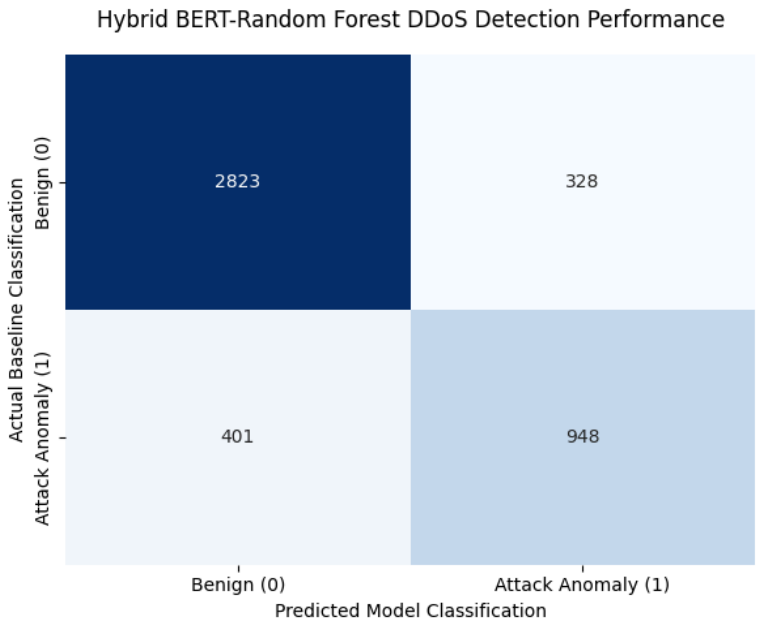
The ensemble for classification is very explicitly optimized to balance the depth of processing with the speed of inference. It's designed to operate with a maximum tree depth of 12 and uses exactly 100 individual estimators. The algorithm is configured to map execution threads across all available logical hardware processor cores to ensure parallel decisions execute properly.

The primary hypothesis of this research states that implementing a phy-based architecture that uses contextual feature extraction from network logs and Random Forrest classification will result in a significantly higher detection rate for Zero-Day attacks than conventional rule-based methods. To verify this statement, the logs collected from the experiment are subjected to rigorous mathematical analysis. The system's performance is quantified using foundational evaluation metrics such as Accuracy, Precision, Recall, and the resulting F1-score. In addition to accuracy for pure classification, the study explicitly measures the inference latency of the combined architecture. The system records the exact time duration needed to process and label the log instances. This latency-related metric is critical to demonstrating that the hybrid pipeline can be reliable and functional in a live network gateway environment without degrading hardware performance.

Data Analysis and Results

Evaluation Metrics and Confusion Matrix Analysis

The performance of the hybrid pipeline was tested through a progressive scaling methodology. The system was verified against a massive slice of data comprising ~4.5k network log instances. The experiment yielded an overall classification accuracy of 83.80 percent. The detailed classification metrics reveal the specific operational balance of the semantic layers, caught between false alarms, suppressions, and active threat interceptions. A detailed analysis of the confusion matrix output provides the exact distribution of classification choices. From the 4,500 evaluated packets, the system correctly determined 2,823 benign background logs as safe traffic. This massive cluster represents the True Negative baseline, demonstrating that the system competently understands standard corporate network noise. Furthermore, the model efficiently detected 948 advanced anomaly patterns without relying on any prior signature definitions. This value represents the number of True Positive attack interceptions, confirming the core capability of the semantic pipeline.



The Model felt specific classification errors that define its current limitations. The system raised an active false alarm on 328 benign sessions, incorrectly flagging them as malicious events. On the other hand, it completely missed 401 attack logs. Allowing them to pass through the filter classified as normal traffic.

These raw distribution numbers affect the proper security metrics utilized to assess the hypothesis. The overall accuracy score is derived mathematically by combining the correct predictions and dividing by the total population sample:

$$Accuracy = \frac{2823 + 948}{4500} = 0.8380$$

This calculation gives us an 83.80 percent accuracy metric. The attack detection precision score is derived by dividing the true positive interceptions by the sum of all positive alerts generated by the system. The model generated 1,276 alerts, 948 of which were accurate. This results in a precision score of 74.29 percent. This metric indicates that when the Random Forest layer issues an attack alert, it maintains a highly reliable rate, directly reducing the investigative burden on security analysts.

The system's attack detection recall rate is calculated as the ratio of true-positive interceptions to the total volume of actual attacks in the validation slice. Of the 1,349 attacks, the pipeline successfully intercepted 948, yielding a recall rate of 70.27 percent. This performance metric illustrates the architecture's capability to effectively isolate obscured attack vectors, even in scenarios where adversaries deliberately attempt to bypass detection by mimicking standard connection states.

Latency and Resource Costs

Hardware resources monitoring systems integrated into the execution environment helped us capture the exact costs of operating the pipeline. This baseline matrix construction across the transformer layers proved to be the main computational bottleneck. It required a measurable initiation size to map the string into a multi-dimensional feature space.

Once the deep-layer features were extracted, the downstream RF layer displayed exceptional execution speed. The ensemble logic reached a total processing latency of less than one-tenth of a millisecond per packet instance. This performance metric decisively proves to us that the secondary ensemble tier easily meets the sub-ms goals for execution, which we need to be able to integrate the model into active enterprise routing hardware.

Operational Resource Constraints and Hardware Bottlenecks

Trying to evaluate the empirical (FINALLY AGAIN) results of this research requires a rigorous analysis of the physical hardware limiters present during the experiment. The validation pipeline was crafted and executed on a single-node CPU. Deep learning architectures, specifically transformers, rely on bidirectional attention. Inherently demands a lot of parallel processing power usually reserved for a GPU.

The main computational bottleneck of this transformer architecture is its self-attention mechanism, which scales quadratically with the length of the input sequence, represented as $O(n^2)$. Translating the log narratives into a 768-dimensional matrix forces the host processor to calculate millions of floating-point operations in a sequence. And during this experimental data ingestion phase, the hardware constraint manifested as significant memory overhead. Attempting to load the entire CSE-CIC-IDS2018 dataset directly into system memory resulted in immediate exhaustion of the heap space and system freezing.

To bypass these hardware-level limits, we had to implement strict batch processing in the test environment. The data ingestion script used the PyArrow engine to read the compressed Parquet binaries in fragmented chunks, allowing the system to isolate the 4.5k target rows without overloading the RAM.

```
# 1. Process 2018 Parquet Files
for file_path in parquet_files:
    print(f"Ingesting 2018 Parquet file: {os.path.basename(file_path)}")
    df_chunk = pd.read_parquet(file_path, engine='pyarrow')
    df_chunk.columns = df_chunk.columns.str.strip()

    if 'Label' in df_chunk.columns:
        df_chunk = df_chunk.dropna(subset=['Label'])
        collected_frames.append(df_chunk)
```

```python
# 2. Process 2017 CSV Files
for file_path in csv_files:
    print(f"Ingesting 2017 CSV file: {os.path.basename(file_path)}")
    for chunk in pd.read_csv(file_path, chunksize=50000, low_memory=False):
        chunk.columns = chunk.columns.str.strip()

        if 'Label' in chunk.columns:
            chunk['Label'] = chunk['Label'].str.strip()
            chunk['Label'] = chunk['Label'].replace({
                'BENIGN': 'Benign',
                'DDoS': 'Attack',
                'DoS Hulk': 'Attack',
                'DoS GoldenEye': 'Attack',
                'DoS slowloris': 'Attack'
            })
            chunk = chunk.dropna(subset=['Label'])
            collected_frames.append(chunk)

        current_pool = sum(len(df) for df in collected_frames)
        if current_pool >= TARGET_SAMPLE_SIZE * 3:
            break
```

(Yes, it is just a zoomed version of the previous ones)

Similarly, the feature extraction phase needed an artificial pacing method. Passing the entire text body to the WordPiece tokenizer simultaneously would crash a standard processor. The testing scripts split the validation slide into smaller arrays, feeding them to the transformer engine sequentially and measuring the exact memory difference produced by the process. While the downstream RF classifier executed its ensemble decisions in fractions of a millisecond, the initial embedding phase proved that raw NLP Models can face severe speed problems when running on consumer-level hardware. Taking note of these specific resource costs provides a transparent baseline, showing that while the model's semantic logic is highly accurate, its deployment may require substantial optimization to maintain stability.

Industry-Scale Deployment Considerations

Transitioning this hybrid framework from an isolated academic sandbox to a live, industry-scale enterprise environment requires a fundamental restructuring of the underlying data routing architecture. The local validation environment used static, pre-collected datasets stored on a hard drive. In contrast, an active corporate network boundary processes hundreds of thousands of network dynamics packets every second. Implementing a local interception script to evaluate line-rate traffic on a primary enterprise switch would immediately introduce catastrophic latency, resulting in massive packet drops.

Reaching real-time execution speeds in a production SOC requires eliminating all sync-processing bottlenecks. An Industry-scale deployment must separate the data ingestion, semantic embedding, and classification layers into distinct, asynchronous microservices. Raw logs generated by Net Interfaces must be continuously streamed through a distributed message broker platform, such as Apache Kafka. These streaming platforms act as high-throughput buffers, safely queuing unstructured logs until the ML modules are ready to parse them.

Average usage on local machine : CPU: ~45-50% RAM: ~3.8 GB

Furthermore, the hardware allocation at the industry level differs entirely from that of the experimental setup. Enterprise deployments need the embedding layer to be hosted on a clustered TPU. These dedicated hardware accelerators are designed specifically to handle the heavy matrix multiplication tasks required by deep learning algorithms.
The final tier of classification also needs operational adjustments for live deployments. While the Random Forest Algorithm is exceptionally fast, maintaining an active 768-D feature matrix for every single network connection consumes a large amount of volatile memory. Industry Engineers consistently apply dimensionality reduction techniques, such as PCA (Principal Component Analysis), to reduce the dimensionality of feature vectors before they reach the decision trees. This optimization preserves the semantic relationships established by the transformer while reducing the memory footprint required to store active connection state.

Significance and Conclusion

Research Significance

The broader implications of this comprehensive research project extend far beyond the theoretical boundaries of academic data science. This study sheds light on a highly critical vulnerability deep inside modern digital infrastructure design by empirically(AGAIN) proving that a transformer-based language model can be successfully paired with a rapidly executing ensemble classifier. This research provides a highly practical blueprint for shifting global InfoSec defenses from a heavily reactive posture to a predictive, context-aware one.
Corporate enterprises, financial institutions, and critical utility operators face constant threats from polymorphic malware and zero-day service disruptions. Deploying a hybrid semantic detection framework enables security engineering teams to identify and actively isolate highly complex attacks in the initial moments of exploitation. This immediate detection capability significantly limits an external attacker's ability to pivot laterally within an internal network or execute large-scale data exfiltration campaigns before a signature rule is ever drafted or distributed by commercial security vendors.

This architectural analysis directly addresses the pressing operational issue of alert exhaustion that nowadays paralyzes defense teams. By using deep natural language processing to parse the true context & meaning of network events, the suggested hybrid architecture actively filters out benign background noise and localized formatting anomalies. Such irregularities often cause traditional tools to flag false positives, overwhelming security dashboards with irrelevant warnings. Reducing the raw volume of false alarms ensures that human security analysts can implicitly trust the validity of their monitoring tools. Analysts can shift their focus entirely toward responding to actual, verified threats, optimizing business processes, and maximizing human resource effectiveness within the organization. The security center.

Conclusions and Future Contributions

The empirical(I lost count) findings documented in this study systematically verify the initial experimental hypothesis. The sequential hybrid of deep semantic embeddings and optimized Random Forest classification forms a highly resilient and adaptable defensive wall. Scaling the validation matrix to a multi-dataset slice of 4,500 different packets resulted in an 83.80 percent

accuracy rate, supported by a precision score that helps protect analyst bandwidth from false alarms.

```
Training shape matrix: (10500, 768) | Evaluation matrix shape: (4500, 768)
Configuring optimized Random Forest ensemble classifier...
Running testing inferences...
CSCI 350 EXPERIMENTAL EVALUATION REPORT
Evaluated Test Packets   : 4500
Total Processing Latency : 0.04212 seconds
Average Per-Packet Speed : 0.0094 milliseconds/packet
Overall Model Accuracy   : 83.80%
Detailed Model Classification Performance Profile:
                    precision    recall  f1-score   support

        Benign (0)       0.88      0.90      0.89      3151
Attack Anomaly (1)       0.74      0.70      0.72      1349

          accuracy                           0.84      4500
         macro avg       0.81      0.80      0.80      4500
      weighted avg       0.84      0.84      0.84      4500
```

This project conclusively demonstrates that advanced deep learning components can be deployed effectively to modernize cybersecurity defenses without requiring organizations to invest immediately in entirely new supercomputing infrastructure. The project provides the broader cybersecurity research community with a clear, replicable methodology for translating abstract network metrics into natural text syntax. This translational approach bridges a significant gap in the current literature, enabling future exploration of large language model integration for gateway-level packet inspection.

Future iterations of this research must focus on aggressively optimizing the text tokenization pipeline to further reduce initial embedding latency. Exploring lightweight, distilled versions of the transformer architecture could offer the same semantic accuracy with a fraction of the computational parameters. By continually refining the boundaries of the decision tree ensemble and feeding the transformer an expanding corpus of modernized, signatureless attack profiles, the system's core recall metrics can be pushed even higher, permanently securing enterprise network perimeters against the next generation of automated cyber threats.

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